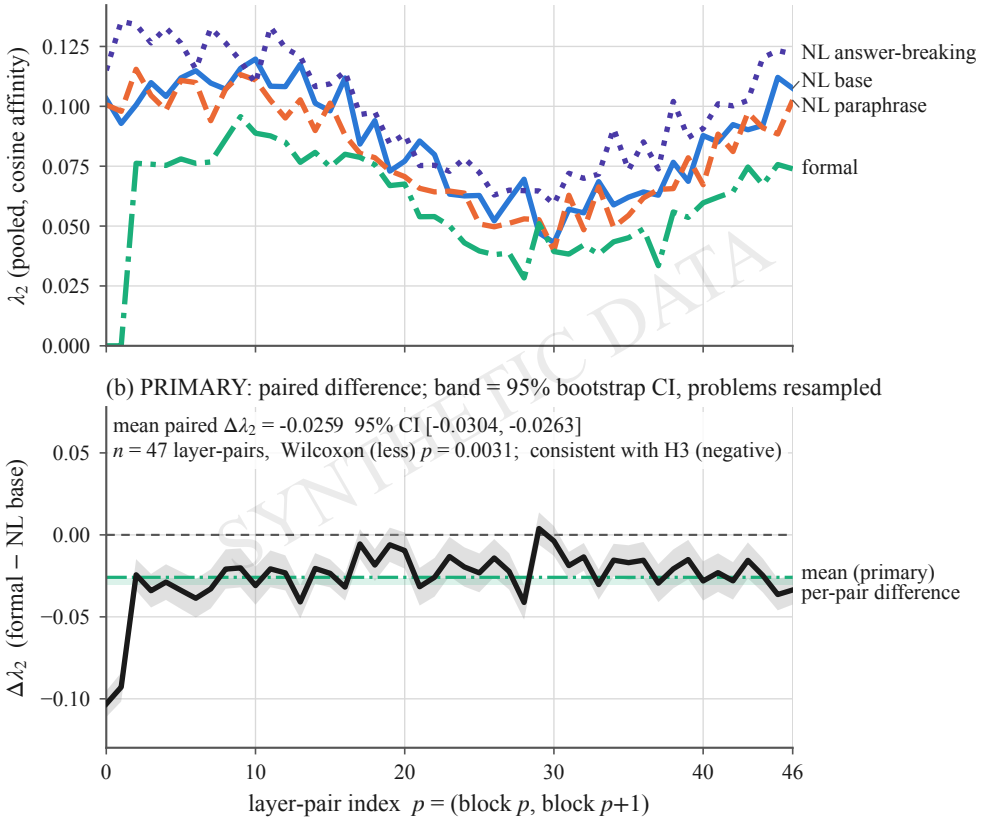
(a) algebraic connectivity by layer-pair, per condition
λ_2 (pooled, cosine affinity)
0.125
0.100
0.075
0.050
0.025
0.000
NL answer-breaking
NL base
NL paraphrase
formal
(b) PRIMARY: paired difference; band = 95% bootstrap CI, problems resampled
mean paired $\Delta\lambda_2$ = -0.0259 95% CI [-0.0304, -0.0263]
n = 47 layer-pairs, Wilcoxon (less) p = 0.0031; consistent with H3 (negative)
$\Delta\lambda_2$ (formal − NL base)
0.05
0.00
−0.05
−0.10
mean (primary) per-pair difference
0
10
20
30
40
46
layer-pair index p = (block p, block p+1)
SYNTHETIC DATA